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Section:	S18 GROUP 8	Grade:	

9.0 User Administration and File Permission

The Ubuntu system allows multiple users to login at the same time since it is a multi-user system. The process of creating, removing and modifying of user accounts in the system is called user administration or user management. Aside from user accounts, the Linux system allows "grouping" of users as a logical collection of user accounts called groups. Same as user accounts, group account can also be created, deleted or modified. The commands that can be used to create accounts and groups are '`adduser`' and '`addgroup`'.

File permissions can also be assigned to both user and group account. File permissions for users or group account can be changed using the '`chgrp`' and '`chmod`' commands.

9.1. User Administration

The Ubuntu Linux system is a multi-user system. User accounts can be created, removed or change. Aside from user accounts, grouping of users can also be made to allow group of users to have access to a service instead to configuring it one at a time.

Step 1 Login to your system and at the command prompt, open the file '`/etc/passwd`'. Examine the file.

1a. What is the file? What does it do? You may research what is this file on the Internet.

```
administrator@lbysyad-s18-20162:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
  :x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
  il:x:8:8:mail:/var/mail:/usr/sbin/nologin
  ws:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
_apt:x:100:65534::/nonexistent:/usr/sbin/nologin
systemd-network:x:101:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:102:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
messagebus:x:103:104::/nonexistent:/usr/sbin/nologin
systemd-timesync:x:104:105:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin
pollinate:x:105:1::/var/cache/pollinate:/bin/false
sshd:x:106:65534::/run/sshd:/usr/sbin/nologin
syslog:x:107:113::/home/syslog:/usr/sbin/nologin
uuidd:x:108:114::/run/uuidd:/usr/sbin/nologin
tcpdump:x:109:115::/nonexistent:/usr/sbin/nologin
tss:x:110:116:TPM software stack,,,:/var/lib/tpm:/bin/false
landscape:x:111:117::/var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:112:118:fwupd-refresh user,,,:/run/systemd:/usr/sbin/nologin
usbmux:x:113:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
administrator:x:1000:1000:SAMARC:/home/administrator:/bin/bash
lxd:x:999:100::/var/snap/lxd/common/lxd:/bin/false
user02:x:1001:1001:,,,:/home/user02:/bin/bash
user03:x:1002:1002:,,,:/home/user03:/bin/bash
administrator@lbysyad-s18-20162:~$
```

It is a plain-text configuration file that stores essential user account information. It is colon separated and each line refers to one user account

Step 2 At the command prompt, open the file '`/etc/group`'. Examine the file.

1a. What is the file? What does it do? You may research what is this file on the Internet.

```
man:x:12:
proxy:x:13:
kmem:x:15:
dialout:x:20:
fax:x:21:
voice:x:22:
cdrom:x:24:administrator
floppy:x:25:
tape:x:26:
sudo:x:27:administrator
audio:x:29:
dip:x:30:administrator
www-data:x:33:
backup:x:34:
operator:x:37:
list:x:38:
irc:x:39:
src:x:40:
gnats:x:41:
shadow:x:42:
mp:x:43:
deo:x:44:
sl:x:45:
plugdev:x:46:administrator
staff:x:50:
games:x:60:
users:x:100:
nogroup:x:65534:
systemd-journal:x:101:
systemd-network:x:102:
systemd-resolve:x:103:
messagebus:x:104:
systemd-timesync:x:105:
input:x:106:
sgx:x:107:
kvm:x:108:
render:x:109:
lxd:x:110:administrator
_ssh:x:111:
crontab:x:112:
syslog:x:113:
uuidd:x:114:
tcpdump:x:115:
tss:x:116:
landscape:x:117:
fwupd-refresh:x:118:
administrator:x:1000:
user02:x:1001:
user03:x:1002:
```

It is plaintext file that shows the groups of the system and which user belongs to them. Its purpose is essential for organizing user permissions for files, directories, etc.

Step 2 At the command prompt type in the command '`sudo adduser student01`'. When prompted for the password, type in '`student123`' after this, you will be prompted for full name, room number, etc. Enter any information about the user account when prompted.

2a. After adding the user, add another user called '`student02`', '`faculty01`' and '`faculty02`'. What is the command you used? How many users are added?

The commands are 'sudo adduser student02', 'sudo adduser faculty01', and 'faculty02'. In total, four users were added

```
Enter the new value, or press ENTER for the default
        Full Name []: Marc Quizon
        Room Number []: 213
        Work Phone []: +63
        Home Phone []: +63
        Other []:
Is the information correct? [Y/n] y
administrator@lbysyad-s18-20162:~$ sudo adduser student02
Adding user `student02' ...
Adding new group `student02' (1004) ...
Adding new user `student02' (1004) with group `student02' ...
Creating home directory `/home/student02' ...
Copying files from `/etc/skel' ...
w password:
type new password:
sswd: password updated successfully
Changing the user information for student02
Enter the new value, or press ENTER for the default
        Full Name []: Sarah Khan
        Room Number []: 213
        Work Phone []: +63
        Home Phone []: +63
        Other []:
Is the information correct? [Y/n] y
```

```
sswd: password updated successfully
Changing the user information for faculty01
Enter the new value, or press ENTER for the default
        Full Name []: Gregory Cu
        Room Number []: 213\
        Work Phone []: +63
        Home Phone []: +63
        Other []:
Is the information correct? [Y/n] y
administrator@lbysyad-s18-20162:~$ sudo adduser faculty02
Adding user `faculty02' ...
Adding new group `faculty02' (1006) ...
Adding new user `faculty02' (1006) with group `faculty02' ...
Creating home directory `/home/faculty02' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for faculty02
Enter the new value, or press ENTER for the default
        Full Name []: Aris Pulumbarit
        Room Number []: 403
        Work Phone []: +63
        Home Phone []: +63
        Other []:
Is the information correct? [Y/n] y
administrator@lbysyad-s18-20162:~$
```

2b. Examine the '`/home`' directory? Were there new directories created matching the date today? What are these directories? Who are the owners of these directories?

Yes new directories were created. These are student01, student02, faculty01, and faculty02. These are the home directories of the new users, and the owner is the user account themself

```
administrator@lbysyad-s18-20162:~$ ls -l /home
total 40
drwxr-x--- 5 administrator administrator 4096 Feb 23 22:15 administrator
drwxr-xr-x 2 root          root          4096 Feb 13 13:21 data
drwxr-xr-x 2 root          root          4096 Feb 13 13:46 data1
drwxr-x--- 2 faculty01     faculty01     4096 Feb 24 13:41 faculty01
drwxr-x--- 2 faculty02     faculty02     4096 Feb 24 13:42 faculty02
drwxrwxrwx 2 root          root          4096 Feb 20 13:34 out
drwxr-x--- 2 student01     student01     4096 Feb 24 13:38 student01
drwxr-x--- 2 student02     student02     4096 Feb 24 13:41 student02
drwxr-x--- 3 user02        user02        4096 Feb 13 13:35 user02
drwxr-x--- 3 user03        user03        4096 Feb 13 13:45 user03
administrator@lbysyad-s18-20162:~$ _
```

2c. Examine the file '`/etc/passwd`'. Were there any changes from last time you examined it? What are the changes? What does each entry separated by '`:`' (colon) means?

Yes the main change is that the accounts we created are included now. Each entry refers to: Username, Password (often 'x' representing shadow), User ID (UID), Group ID (GID), GECOS (user info), Home Directory, and Login Shell.

```
student01:x:1003:1003:Marc Quizon,213,+63,+63:/home/student01:/bin/bash
student02:x:1004:1004:Sarah Khan,213,+63,+63:/home/student02:/bin/bash
faculty01:x:1005:1005:Gregory Cu,213\,+63,+63:/home/faculty01:/bin/bash
faculty02:x:1006:1006:Aris Pulumbarit,403,+63,+63:/home/faculty02:/bin/bash
administrator@lbysyad-s18-20162:~$
```

Step 3 At the command prompt, type in the command '`sudo addgroup faculty`' and '`sudo addgroup student`'.

3a. What does the command do?

```
administrator@lbysyad-s18-20162:~$ sudo addgroup faculty
Adding group `faculty' (GID 1007) ...
Done.
administrator@lbysyad-s18-20162:~$ sudo addgroup student
Adding group `student' (GID 1008) ...
Done.
administrator@lbysyad-s18-20162:~$
```

These commands add new groups. In this case, faculty and student was added.

3b. Examine the file '`/etc/group`'. Were there any changes from the last time you examined it? What are the changes?

```
student01:x:1003:
student02:x:1004:
faculty01:x:1005:
faculty02:x:1006:
faculty:x:1007:
student:x:1008:
administrator@lbysyad-s18-20162:~$
```

Yes, the /etc/group shows that new groups that were made and their GIDs.

Step 4 At the command prompt, type in the command '`sudo addgroup student01 student`'.

4a. What does the command do?

This command adds the username student01 into the students group

```
administrator@lbysyad-s18-20162:~$ sudo addgroup student01 student
Adding user `student01' to group `student' ...
Adding user student01 to group student
Done.
administrator@lbysyad-s18-20162:~$ cat
```

4b. Examine the file '`/etc/group`'. Were there any changes from last time you examined it? What are the changes?

```
student:x:1008:student01
administrator@lbysyad-s18-20162:~$
```

There is a new line entry which states student01 is part of the students group

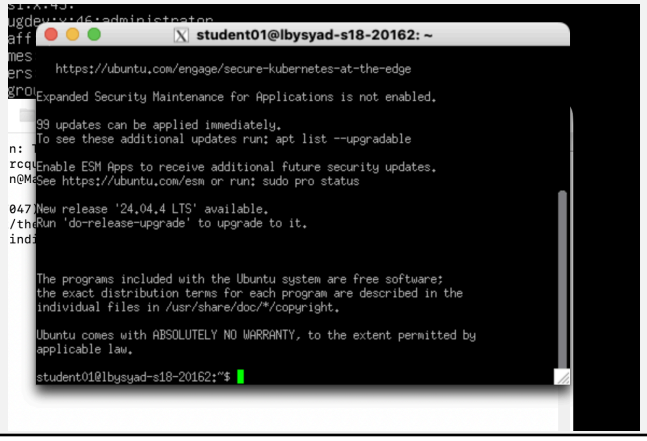
4c. Add user account '`faculty01`' to the group faculty. What command did you use?

'sudo addgroup faculty01 faculty'

```
administrator@lbysyad-s18-20162:~$ sudo addgroup faculty01 faculty
Adding user `faculty01' to group `faculty' ...
Adding user faculty01 to group faculty
Done.
administrator@lbysyad-s18-20162:~$
```

Step 5 On another machine, use SSH to login to your system using the user account '`student01`' account.

5a. Use the password you set during account creation to login. Are you able to login? Exit the account.



Yes we were able to login using student123

5b. Login using the default (administrator) and type in the command '`sudo passwd student01`'. Use SSH to login to your system using the user account '`student01`' using the previous password. Are you successful? Why or why not?

We changed the password to student12345. Hence, it was not successful using the previous password (student123) because the password changed.

5c. Use the password you set using the '`passwd`' command earlier? Are you able to login? Why or why not?

Yes. When I used student12345, I was able to login successfully

5d. What does the '`passwd`' command do?

The 'passwd' command is used to change or update a password for a user account in the system that is used when logging in.

Step 6 At the command prompt, type in the command '`sudo chage -l student01`'.

6a. What information does the command display?

The command showed the password change history and account age information

```
administrator@lbysyad-s18-20162:~$ sudo chage -l student01
Last password change                                    : Feb 24, 2026
Password expires                                        : never
Password inactive                                       : never
Account expires                                         : never
Minimum number of days between password change          : 0
Maximum number of days between password change          : 99999
Number of days of warning before password expires       : 7
```

6b. At the command prompt, type in the command '`sudo chage student01`'. Change the "Minimum password age" to 1, "Maximum password age" to 3 and an "Account expiration date" of 7 days from now. Use the default values for the other options.

6c. Use SSH to login to your system using the student01 user account. At the command prompt change your password using the '`passwd`' command. Were you able to change the password?

```
tudent01@lbysyad-s18-20162:~$ passwd
hanging password for student01.
urrent password:
ou must wait longer to change your password.
asswd: Authentication token manipulation error
```

No. The password was unchanged. This is because the minimum password age is 1 day. 1 day has not passed hence the password cannot be changed

6d. At the command prompt, change the password again. Were able to change the password? Why or why not? Logout after this.

```
student01@lbysyad-s18-20162:~$ passwd
Changing password for student01.
Current password:
You must wait longer to change your password.
passwd: Authentication token manipulation error
passwd: password unchanged
student01@lbysyad-s18-20162:~$
```

No, it still shows the same results

6e. Change the date of the system to the password expiration day of user account '`student01`' and login using SSH. Was there a warning to change the password? What does it say?

Yes there was a warning to change the password

```
login as: student01
student01@ccscloud.dlsu.edu.ph's password:
Pre-authentication banner message from server:
| Warning: your password will expire in 3 days.
End of banner message from server
Warning: your password will expire in 3 days.
```

Note: Turn-off time synchronization first before changing the date. sudo timedatectl set-ntp off

6f. Logout but do change the password. Change the system date to one day after the password expiration day of user account '`student01`' and use SSH to login. Were you able to login? Was there a warning? What was the message?

> Yes there is still a warning, this time it says that our password will expire in 0 days

6g. Logout but do change the password. Change the system date to two days after the password expiration day of user account '`student01`' and use SSH to login. Were you able to login? Why or why not?

> Not exactly, it prompted the user to change their password immediately. This is because the password has expired

6h. Change the system date back to the correct date.

Step 7 At the command prompt, type in the command '`sudo deluser student01`'.

7a. What does the command do?

> This command deletes the user 'student01'

7b. Examine the '`passwd`' file. Does the user account '`student01`' still exist? Why or why not?

> By using 'cat /etc/passwd', we can see that the student01 entry is still there. It still exists. Wherein, the home directory and files remain but the login and system entries are gone

7c. Check the '`/home`' directory. Does the home directory of user '`student01`' still exist? Does the command clear or remove the directory?

> Yes, using the ls command, we can see that it still exists. The 'deluser' command does not remove the directory of that user, only the user account.

7d. Exit the system.

9.2. File Permission – Using chmod, chown, and chgrp

File permission or access rights allows a system administrator to tailor fit access of files on the server. File permission allows the system administrator to set accesses of certain directories or files to specific users or group of users.

Step 8 Login to your system and create a directory named '`student_data1`' at the root (`/`) directory.

8a. At the command prompt type in the command '`ls -ld /student_data1`'. Who are the owner and group owner of the directory?

> The owner and group owner of the directory is both root

```
administrator@lbysyad-s18-20162:/$ sudo mkdir student_data1
administrator@lbysyad-s18-20162:/$ ls -ld /student_data1
drwxr-xr-x 2 root root 4096 Feb 24 21:27 /student_data1
```

8b. At the command prompt type in the command '`sudo chgrp student /student_data1`'. Did the group owner of the file become different? Who is the new group owner of the directory?

Yes, it changed. The group is now students

```
drwxr-xr-x 2 root root 4096 Feb 24 21:27 /student_data1
administrator@lbysyad-s18-20162:/$ sudo chgrp student /student_data1
administrator@lbysyad-s18-20162:/$ ls -ld /student_data1
drwxr-xr-x 2 root student 4096 Feb 24 21:27 /student_data1
administrator@lbysyad-s18-20162:/$ sudo chown student02 /s_
```

8c. At the command prompt type in the command '`sudo chown student02 /student_data1`' and '`sudo chmod 775 /student_data1`'. Did the owner of the file become different? Who is the new owner of the directory? What are the permissions of the owner, group and others?

The owner of the file is now student02 and the permissions now allow rwx access for the "owner' and 'group'. While, 'others' only has rx access.

Note: To allow other users or groups an access to read only (browsing) to a directory, there should be an "execute" permission.

8d. At the command prompt type in the command '`cd /student_data1`' and create an empty file using the command '`touch file1`'. Are you able to create the file why?

No, permissions were denied. Because we are an administrator which is considered the 'others' group (only has rx access). Hence, we cannot create the file

8e. At the command prompt type in the command '`sudo touch file1`'. Are you able to create the directory? Why or why not?

file1 is not a directory because we used the 'touch' command. This is a file and we were able to create it successfully due to the 'sudo' command

8f. Login remotely using the '`student02`' account via SSH. Go to the '`/student_data1`' directory and type in the command '`touch file2`'. Are you able to create the file? Why or why not? What is the default permission of the file for owner, group and others? (Do not exit the SSH connection)

Yes, the student02 user account was able to create the file02 as student02 is part of the ' ' (has rwx access). Hence, they can create new files.

8g. Using the `administrator` account, create the user account '`student03`'. Add the user account '`student03`' to the '`student`' group after creating the user account. What is the command to create the user account and adding it to the student group?

The commands are ' sudo adduser student03' and 'sudo addgroup studetn03 student'/

8h. Create another SSH connection but login as '`student03`'. Go to the '`/student_data1`' directory and type in the command '`touch file3`'. Are you able to create the file? Why or why not?

Yes it is successful because student03 is part of the student group. They have rwx access to the file

8i. Using the '`student03`' SSH connection, edit the file '`file2`' and enter the text "`The quick brown fox jumps over the lazy dog`." and try to save it. Are you able to save or modify it? Why or why not?

```
student03@lbysyad-s18-20162:/student_data1$ nano file2
student03@lbysyad-s18-20162:/student_data1$ ls -ld file2
-rw-rw-r-- 1 student02 student02 0 Feb 24 21:33 file2
student03@lbysyad-s18-20162:/student_data1$
```

Here are the permissions of file2. We were not able to modify it because the owner and group is student02. student03 is considered 'others' and they only have r access

8j. Using the '`student02`' SSH connection, edit the file '`file2`' by entering the text "`The quick brown fox`". Are you able to edit it? Why or why not?

Yes, we are able to edit the file because we are in student02's account. Which has rw access.

8k. Using the '`student02`' SSH connection, change the permission of file '`file2`' by using the command '`chmod 765 file2`'. Are you able to change the permission? Why or why not? What is the new permission?

```
student02@lbysyad-s18-20162:/student_data1$ chmod 765 file2
student02@lbysyad-s18-20162:/student_data1$ ls -ld file2
-rwxrw-r-x 1 student02 student02 20 Feb 24 21:41 file2
```

We are able to change the permission because student02 has rw access. Now, the new permissions are rwx (owner), rw (group), rx (others).

8l. Using the '`student03`' SSH connection, try to edit the file '`file3`' by entering the text "`The quick brown fox`" and save the file. Are you able to save or modify it? Why or why not?

Yes, we were able to edit and save the file3 as the privileges changed. Because student03 (others) has rx access

8m. Using the '`student03`' SSH connection, delete the file '`file2`'. Are you able to delete it? Why or why not?

Yes. This is because we have rx access

Step 9 Using the `administrator` account, create another directory named '`faculty_data1`' inside the root (`/`) directory. Change the group ownership to faculty. Make sure that the group permission allows the group to have a read, write and execute access to it. For the permission of other users, make sure that they only have read access.

9a. Write down the series to command you used to be able to accomplish the requirement.

sudo chgrp faculty /faculty_data1
chmod 774 faculty_data1

9b. Creating a new SSH connection, login as '`faculty01`' and go to directory '`/faculty_data1`'. Type in the command '`touch facultyfile1`'. Are you able to create the file why?

```
faculty01@lbysyad-s18-20162:/faculty_data1$ touch facultyfile1
faculty01@lbysyad-s18-20162:/faculty_data1$ ls
facultyfile1
faculty01@lbysyad-s18-20162:/faculty_data1$ ls -ld facultyfile1
-rw-rw-r-- 1 faculty01 faculty01 0 Feb 24 21:54 facultyfile1
```

Yes it worked. This is because faculty01 is part of the faculty group. Which has rwx access.

9c. Using the '`student02`' SSH connection, try to create a file in the '`faculty_data1`' directory named '`facultyfile2`'. Are you able to create it? Why or why not?

No, student02 is part of the 'others' for faculty_data1 which only has r access. Hence, they cannot create files

9d. Change the permission of the directory '`faculty_data1`' to read, write access for other users. Use the '`student02`' connection to create the '`facultyfile2`' file. Are you able to create it? Why or why not?

No, the permission is denied to create the file. To do these commands, we would need rwx access.

Step 10 Using the '`student02`' SSH connection, create the file '`file4`' using the command '`touch file4`' in the '`/student_data1`' directory. Change the permission of the '`/student_data1`' directory by using the command '`chmod 1777 /student_data1`'. This command sets the sticky bit on the file. Also change the permission of the '`file4`' by using the command '`chmod 777 file4`'.

10a. Using the '`student03`' connection, try to delete the file. Are you able to delete it?

No, we are not permitted to delete it. The '1' refers to sticky bit which only permits the owner to delete and rename the file

10b. Using the '`student02`' connection, try to delete the file. Are you able to delete it? What does the sticky bit do?

Yes we are able to delete it. The sticky bit enables only the owner to delete/rename files

10c. Exit all accounts including all SSH connections.

9.3. File Permission – Using ACL (Access Control List)

Although the traditional file permission can allow access to certain users and groups, Linux ACL allows the system administrator or even a user, to have finer permission settings. An example use of ACL is allowing two groups to access directories with different directories which traditional permission settings cannot do.

Commands used to set the ACL are '`getfacl`' and '`setfacl`'. The '`getfacl`' command displays the ownership and permission of a directory or file while the '`setfacl`' command sets the permission of a directory or file.

Step 11 Login to the Ubuntu Linux system and add the following user: '`staff01`', and '`staff02`'. Add the user accounts to the group '`staff`'.

11a. Add a directory named '`faculty_data02`' in the root (`/`) directory. Change the group ownership to the group '`faculty`' with the group permission of read and write. Leave the owner of the directory as '`root`'.

11b. At the command prompt, type in the command '`getfacl /faculty_data02`'. Who are the owner and group owner? What are the permissions of the directory?

The owner is the root and the group owner is the faculty. The permissions of the directory is rwx.

```
administrator@lbysyad-s18-20162:/$ sudo getfacl faculty_data02
# file: faculty_data02
# owner: root
# group: faculty
user::rwx
group::rwx
other::rw-
```

11c. Using an SSH connection, login as '`faculty01`'. Go to directory '`/faculty_data2`' and create an empty file called '`faculty_file1`'. Are you able to create the file? Why or why not?

Yes, the user faculty01 was able to create the file as it has rwx permissions.

11d. Using another SSH connection, login as '`student02`' and go to directory '`/faculty_data2`'. Create an empty file called '`student_file1`'. Are you able to create the file? Why or why not?

No, permission denied. This is because student02 only has rw access.

11e. Using another SSH connection, login as '`faculty02`' and go to directory '`/faculty_data2`'. Edit the file called '`faculty_file1`' by entering the "The quick brown fox" and save the file. Are you able to save and modify the file? Why or why not?

Yes, the file is successfully saved and modified. This is because faculty02 has rwx permission

Step 12 At the command prompt, type in the command '`sudo setfacl -m u:faculty02:rw /faculty_data2/faculty_file1`' and '`sudo setfacl -m g:student:rw /faculty_data2`'.

12a. At the command prompt, type in the command '`getfacl faculty_data02/faculty_file1`'. Who are the user and group owners and what are the permissions of the file?

The owner is faculty02 and the group is faculty02. The user permissions is rw, the group permissions is rw, and others is just read permissions.

12b. Using the '`faculty01`' SSH connection, edit the file '`faculty_file1`' by entering the text "`The quick brown fox`" and save the file. Are you able to modify and save the file? Why or why not?

No, we cannot modify or save the file. This is because of 'u:faculty02', this indicates only faculty02 will have rw access.

12c. Using the '`faculty02`' SSH connection, edit the file '`faculty_file1`' by entering the text "`jumps over the lazy dog.`" and save the file. Are you able to modify and save the file? Why or why not?

Yes, we modified and saved the file successfully.

12d. Using the '`student01`' connection, try to create an empty file in the '`/faculty_data2`' directory named '`student_file1`'. Are you able to create the file? Why or why not?

```
student01@lbysyad-s18-20162:~$ cd /faculty_data02
-bash: cd: /faculty_data02: Permission denied
```

No most definitely for surely not. We attempted using 'touch faculty_data02/student_file1' but permission was denied as well. This is because student01 is considered 'others' and they cannot cd into faculty_data02 as we only have rw access.

12e. Using another SSH connection, login as '`staff01`' and go to directory '`/faculty_data2`'. Try to create an empty file named '`staff-file1`'. Are able to create the file? Why or why not?

```
staff01@lbysyad-s18-20162:~$ cd /faculty_data02
-bash: cd: /faculty_data02: Permission denied
```

No, the staff

12f. Using the '`staff01`' SSH connection, try to edit the '`faculty_file1`' file and enter the text "`and the quick cat too`". Are you able to modify and save the file? Why or why not?

No, the user staff01 cannot edit the file because it does not have write permissions for the file. We did not set any ACL for the staff group.

Step 13 At the command prompt, type in the command '`sudo setfacl -x g:student /faculty_data2`'.

13a. Using the '`student01`' SSH connection, try to create an empty file named '`student_file2`'. Are you able to create the file? Why or why not?

No, this is because the ACL permission for the student group was removed

13b. What is the command to remove the ACL permission of user '`student01`' on the file '`faculty_file1`'?

sudo setfacl -x u:student01 /faculty_data02/faculty_file1

13c. Exit or shutdown the system.

9.4. Conclusion/Reflection

What is the purpose of assigning users to groups?

It is more efficient to give and change permissions for each group and for troubleshooting when a group already has its own permissions by default. It becomes useful in organizing as you are splitting it into roles, and for system security.

Provide a scenario when it is more appropriate to use ACLs over generic permissions using `chmod`.

It is more appropriate to use ACLs when we need to be more specific with user and group permissions. For example, we have a student leader in the students group that needs special access to files. We would use an ACL to give privilege to the specific student leader in the students group only. If chmod was used, it is not possible to specifically modify the privileges of a specific member of a group.